PHD FELLOWSHIP STRATEGIC BASIC RESEARCH - PROJECT OUTLINE
Holonic Demanufacturing under Structural Uncertainty: AI Integration through Evidence-Grounded Digital Twins

Rationale and positioning with respect to the state of the art

The transition to a circular economy requires demanufacturing systems capable of operating on end-of-life (EoL) electronics under **high product-model heterogeneity** and **condition-dependent structural variability**, including unknown damage states, diverse fastening strategies, degradation (e.g., adhesives, batteries), and partially observable internal configurations [1]. Although robotic disassembly and recycling have advanced in academic prototypes and industrial pilots [2], broad industrial deployment for **heterogeneous laptop and smartphone** streams remains limited, as many approaches rely on controlled assumptions or product-specific constraints [3]. As a result, disassembly is still largely performed manually by skilled operators [1], creating persistent challenges for scalability and worker safety in e-waste operations [4].

A central difficulty lies in the **epistemic structure of demanufacturing control** (figure 1). Conventional manufacturing assumes a predefined product model (Bill of Materials and routing) and treats uncertainty mainly as parametric disturbance within a fixed structural template [5]. In contrast, demanufacturing involves instance-dependent feasibility: the admissible action set, operational hazard constraints, and recoverable outputs may be only partially observable at intake and are progressively revealed through inspection and partial disassembly [1][6]. This introduces **structural uncertainty** in addition to parametric variability: the *space of feasible operations* evolves as knowledge about product structure and condition is updated [6][7]. Recurring exception cases (e.g., damaged fasteners, adhesive failures, battery swelling, missing parts) materially affect routing and constraint-compliance decisions in real streams [8][9]. Shop-floor control therefore becomes a **coordination-under-uncertainty** problem in which material flow and knowledge acquisition co-evolve, requiring robust and constraint-compliant progress under evolving structural uncertainty rather than globally optimal planning under fixed assumptions.

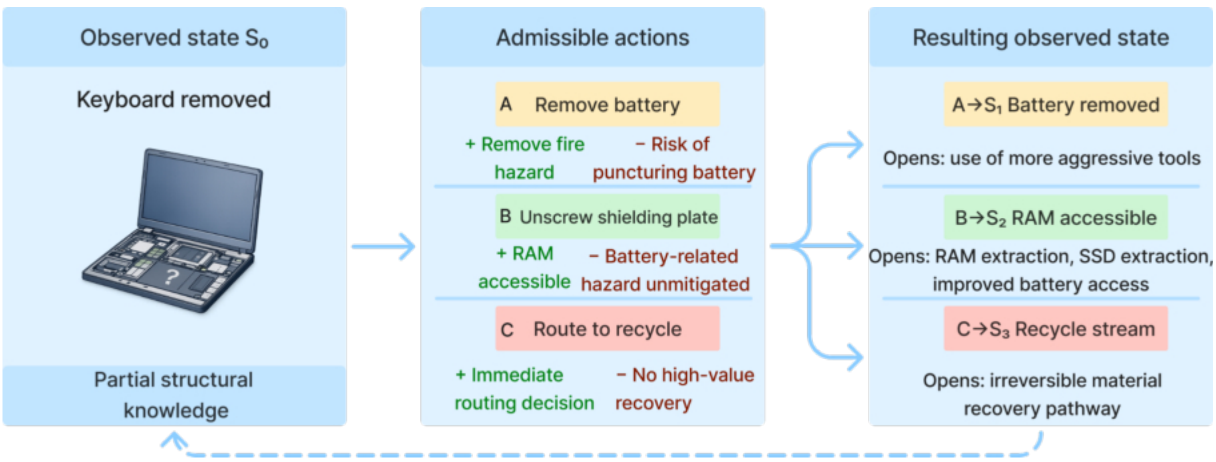

Figure 1. *In demanufacturing, product structure is only partially known at intake. As inspection updates the as-found model, feasible actions and routing decisions evolve accordingly.*

International research addresses aspects of this challenge along three largely separate tracks. First, automated disassembly has advanced perception, task and motion planning, and cell-level automation, yet robustness to heterogeneous EoL variability and exception-heavy operation remains limited [1]. Second, distributed shop-floor control paradigms, particularly holonic and multi-agent architectures such as PROSA and ADACOR, provide scalable coordination through role-based entities (e.g., product, resource, order) and decentralized negotiation [10][11]. However, demanufacturing exception handling often depends on **unstructured, context-rich evidence** (e.g., partial sensor data, ambiguous feedback, incomplete device histories) that is difficult to formalize within predefined coordination logic [12]. Third, Large Language Models (LLMs) demonstrate strong contextual reasoning capabilities and are increasingly explored for manufacturing planning support [13], but their probabilistic and opaque behavior has largely confined them **outside execution-time control loops** due to constraint-enforcement and auditability concerns [14]. Early efforts to combine LLMs with holonic reasoning remain conceptual and do not specify how formal constraint guarantees can be preserved during shop-floor coordination [15].

This project addresses the gap at the intersection of these strands: defining architectural mechanisms that allow semantic reasoning to assist decentralized, execution-time coordination while preserving formal operational and hazard constraints. Probabilistic components may support negotiation and exception handling, but never receive actuator authority. A PROSA-inspired holonic architecture is adopted because its **structural decomposition along real-world entities (product, order, resource)** supports scalable coordination under variability [10][11].

Integration in brownfield environments without compromising certified PLC-based actuation is enabled through an evidence-grounded digital twin that maintains a shared, time-stamped representation of relevant physical and operational states, including uncertainty. The twin is an **event-sourced, provenance-aware mirror**: observations, state updates, and coordination decisions are recorded as immutable traces with freshness and confidence metadata. Inconsistencies between observed and represented state trigger conservative handling (e.g., additional sensing, guarded execution, or escalation), ensuring epistemic alignment between software and physical reality.

This project introduces a **bounded semantic mediation** layer for holonic demanufacturing control. Semantic reasoning supports exception handling by interpreting context-rich evidence, but its probabilistic nature is explicitly contained. The semantic module may propose structured recommendations within a fixed vocabulary of admissible intents defined by the verified coordination model; it cannot introduce new actions, bypass PLC interlocks, or access actuators. All outputs must be trace-linked to digital-twin evidence and pass through a deterministic validation gate before execution proceeds.

A running example clarifies the intended interaction: when a laptop is flagged for a possible swollen battery during cover removal, the digital twin records the observation with evidence pointers and uncertainty metadata; the order holon enters an exception state; the AI module proposes admissible intents such as “request additional sensing,” “route to quarantine,” or “escalate to supervisor”; and the validation gate filters these recommendations against current state and supervisory constraints. In this way, semantic assistance enhances coordination without expanding the reachable state space of the deterministic control layer.

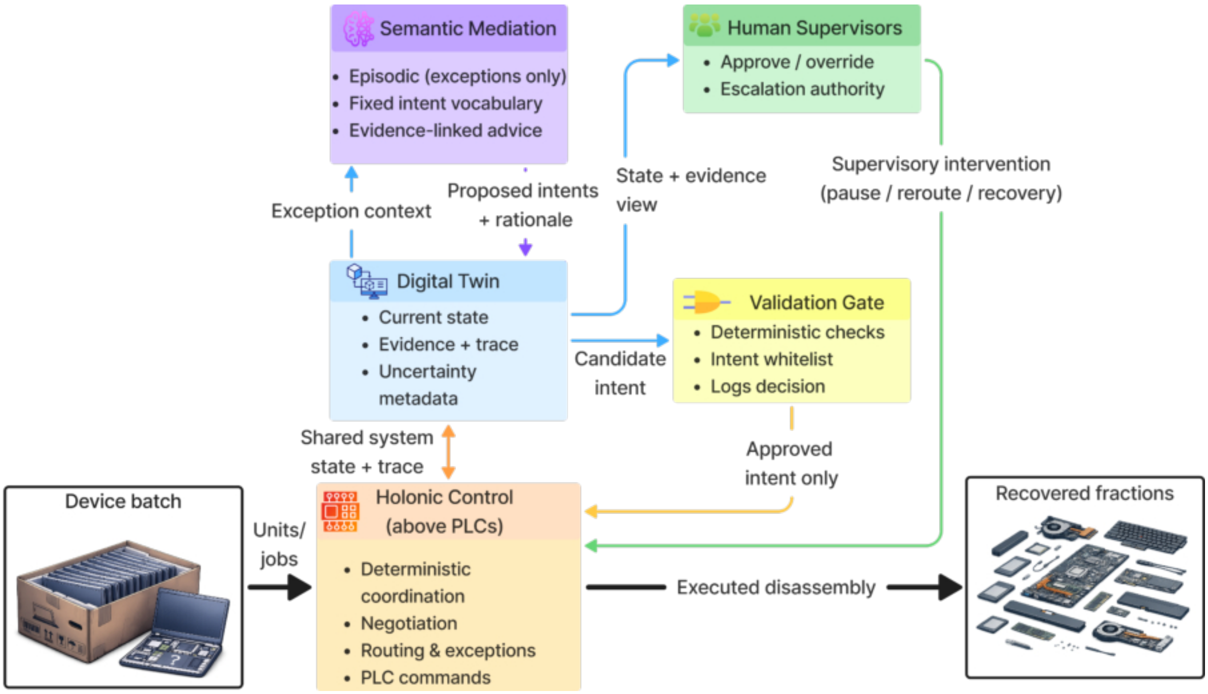

Figure 2. *Layered architecture for adaptive demanufacturing. A PROSA-inspired holonic control layer coordinates PLC-governed equipment, supported by an evidence-grounded digital twin maintaining state and traceability. An episodic semantic mediation layer provides bounded recommendations during exceptions, while AI remains outside the actuator path; execution proceeds only via the deterministic control stack, with human override under predefined escalation conditions.*

The proposed solution (figure 2) instantiates a PROSA-inspired holonic coordination layer above PLC-governed equipment, where **product holons** represent individual EoL units and their evolving condition, **order holons** encode process goals and recovery strategies, and **resource holons** represent stations, tools, and human capabilities; negotiation and disturbance handling occur at this layer [10].
An evidence-grounded digital twin maintains a provenance-aware, time-stamped system state to support coordination and auditability [16][17], while learning mechanisms reduce structural uncertainty without expanding admissible actions.
Above this, a bounded semantic mediation layer (LLM-based) is invoked episodically during exceptions, producing structured recommendations within a fixed intent vocabulary that remain strictly outside the actuator authority path [14]. Human supervision operates as a coordination role with authority to validate, request additional evidence, pause, reroute, or override execution under predefined escalation conditions, ensuring that final actuation authority and constraint enforcement remain external to the AI component and within the deterministic control layer.

Relative to ongoing efforts, the project addresses a complementary layer of the problem. Industrial initiatives such as Apple's Daisy, Hiro Robotics, and Fraunhofer IFF's iDEAR primarily advance cell-level robotic disassembly, perception, and skill execution [3][18][19]. In contrast, this project focuses on the **coordination and information-flow architecture** required to manage structural uncertainty across heterogeneous streams, with explicit operational and hazard constraints and auditable integration of semantic reasoning during execution-time control.
Nationally, the work aligns with Flemish manufacturing innovation agendas (adaptive robotics, human-centric Industry 5.0) and circular-economy initiatives, while addressing a concrete gap in

deployable adaptive automation for heterogeneous e-waste disassembly [20][21][22]. By shifting hazardous manual work toward human-supervised, exception-driven operation supported by auditable coordination and bounded semantic assistance, the project positions Flanders to advance scientifically grounded, constraint-compliant circular manufacturing systems.

Scientific research objectives

This project addresses the main research question: **What architectural mechanisms enable bounded semantic mediation to enhance coordination under structural uncertainty in holonic electronics demanufacturing systems, while preserving formally specified invariants of the discrete-event control layer?**

The core tension lies between progressively revealed structural uncertainty in demanufacturing and the requirement for formally guaranteed coordination in brownfield systems. The project advances three integrated steps: (i) formalizing structural uncertainty in the coordination and digital-twin model, (ii) maintaining an evidence-grounded twin under partial observability through conservative learning, and (iii) integrating LLM-based semantic reasoning through a contract and validation gate that restricts its influence to invariant-preserving decisions.
The scientific contribution is a layered, formally analyzable interface between non-deterministic semantic reasoning and deterministic holonic control, structured through four objectives.

Objective 1: Formalize and model structural uncertainty for holonic demanufacturing coordination
Although prior work recognizes that disassembly operates under progressively revealed structural uncertainty, most experimental setups still assume fixed feasibility structures or complete knowledge, leaving coordination under evolving structural conditions an open challenge [1][8][9][12][23][24].
This objective models structural uncertainty as uncertainty over the realized disassembly feasibility structure (e.g., precedence relations, admissible operations). A parameterized uncertainty-aware discrete-event model is developed in which observations update both state estimates and enabled operations, while supervisory invariants (constraint compliance, liveness, resource bounds) remain preserved.
A representative laptop demanufacturing cell provides the instantiation context. The PLC layer is treated as fixed deterministic control, while a PROSA-inspired holonic coordination layer [10] negotiates and handles exceptions over the uncertainty-aware DES abstraction. Formal links to Petri-net invariant analysis provide guarantees of nonblocking behavior and constraint preservation under structural belief updates [25]. Controlled structural variants define reproducible uncertainty regimes.

Objective 2: Learn evidence-grounded digital-twin updates and information acquisition under partial observability
This objective develops learning mechanisms that maintain alignment between physical system and digital twin under partial observability. Two components are developed and evaluated: (i) *learning-to-update*, inferring latent process-relevant attributes from incomplete evidence; (ii) *learning-to-ask*, selecting targeted sensing or inspection actions that reduce uncertainty sufficiently to proceed within the admissible action space.
The outcome is an uncertainty-aware, learning-enhanced digital-twin framework with quantified performance in reducing unresolved uncertainty and exception escalation, while preserving evidence traceability and conservative fallback guarantees.

Objective 3: Specify a bounded semantic mediation contract for LLM-assisted exception handling
This objective defines a formal contract governing interactions between LLM-based reasoning and invariant-preserving holonic coordination.
The semantic module is invoked only in exception states and produces structured recommendations

within a fixed schema and vocabulary of admissible intents (e.g., additional sensing, compliant rerouting, escalation) [14]. A validation gate enforces consistency with the supervised discrete-event model: no new controllable events may be introduced, PLC interlocks cannot be bypassed, and the reachable state space of the verified closed-loop system cannot expand.
Recommendations must be trace-linked to explicit digital-twin evidence in order to reduce the risk of model hallucinations, and conservative fallback policies handle inconsistency or high uncertainty.
The outcome is a formally specified mediation contract and validation architecture, with property-preservation arguments demonstrating invariant preservation under bounded semantic influence, and a reusable pattern for constraint-preserving LLM integration in holonic control.

Objective 4: Constraint preservation and performance characterization with transferable integration patterns
The final objective formally and empirically characterizes the constraint-preservation and performance envelope of bounded semantic mediation under structural uncertainty.
Formally, operational constraint, liveness, and resource invariants are shown to remain preserved independently of the internal behavior of the semantic module. Empirically, the architecture is evaluated against non-semantic holonic baselines across defined uncertainty regimes, measuring exception resolution, recovery dynamics, throughput impact, and escalation rate.
Sensitivity analyses assess how coordination topology and uncertainty profiles shape performance and guarantees beyond the representative cell.
The outcome is a rigorous characterization of invariant guarantees and operational trade-offs, including quantified comparisons and validated integration guidelines for structurally uncertain holonic demanufacturing systems.

Figure 3. *Demanufacturing lab in KU Leuven that will be used as a testbed in this project: conveyor-based multi-sensor line (left) and robotic manipulation workstation (right) used for autonomous battery-driven product disassembly research.*

Research methodology and work plan
The project develops and validates a **bounded semantic mediation (BSM)** architecture for **holonic coordination of demanufacturing under structural uncertainty**. The experimental context is a **brownfield-inspired laptop demanufacturing cell** (figure 3) in the KU Leuven Re- and Demanufacturing Lab: an industrial robot with interchangeable grippers, conveyor, multi-modal vision/3D/X-ray sensing, and two stations (waterjet cutting; automated unscrewing/drilling for fastener/battery removal). **Low-level PLC/robot sequencing and safety interlocks are treated as fixed**; the research focuses on **orchestration and rerouting above the PLC layer**. A **UE5 digital twin already exists** and will be extended to support this project.
The methodology combines (i) **formal discrete-event modeling and supervisory control** to guarantee operational constraint and liveness invariants under brownfield constraints, (ii) an **event-sourced,**

uncertainty-aware digital twin to represent partial observability and evidence provenance, (iii) **conservative learning** to update beliefs and decide when to acquire information, and (iv) a **deterministic validation gate** that constrains any LLM-derived recommendations to the **supervisor-enabled decision set**, enabling formal assurance that semantic assistance cannot violate invariants. The research steps are organized into five coherent work packages (WPs), with continuous integration into the evaluation environment.

This combination is selected to provide formal assurance while enabling adaptive reasoning. Supervisory discrete-event modeling guarantees invariants under brownfield constraints. The event-sourced twin ensures traceability and reproducibility. Conservative learning reduces uncertainty without expanding admissible actions. Decision containment (via intersection with the supervisor-enabled set) provides stronger guarantees than post-hoc output filtering. Together, these elements form a coherent assurance case for integrating probabilistic reasoning into deterministic industrial control.

WP1: Brownfield benchmark, structural uncertainty models, and evaluation platform

WP1 formalizes the representative cell and produces a reusable benchmark (models + regimes + evaluation loop). In this WP, the **PROSA-inspired holonic architecture is explicitly instantiated above the fixed PLC layer**, defining product, resource, and order holons and their interaction logic for task negotiation, routing, and exception handling. The system is modeled above the PLC layer using a parameterized discrete-event system (DES) capturing routing alternatives, resource constraints, task dependencies, and exception states.

Structural uncertainty (unknown product variants, hidden fasteners, uncertain battery condition, missing part information) is represented as belief sets/uncertainty descriptors attached to the product holon state; observations update these beliefs. Operational constraint, liveness, and resource invariants are encoded as supervisory constraints that define the **supervisor-enabled set of admissible coordination decisions**, ensuring all holonic actions remain compliant with brownfield control limits.

Outputs. A validated evaluation environment (simulation/emulation + instantiated holonic coordination loop) with parameterized uncertainty regimes and baseline coordination policies.

WP2: Uncertainty-aware digital twin state, evidence, and trace layer

WP2 implements the evidence-grounded digital twin introduced in the Rationale. The twin ingests heterogeneous evidence (vision, 3D, X-ray, PLC logs, operator inspections) into a standardized schema with **provenance, freshness, and confidence** metadata. An **event store** enables deterministic replay, auditing, and dataset generation for evaluation and learning. The **UE5 twin** is extended to expose the necessary interfaces (state visualization, trace queries, and replay-driven inspection of runs).

Outputs. Twin state/evidence specifications, event store + replay, baseline conservative belief updates, UE5 integration for visualization and debugging.

WP3: Learning-to-update and learning-to-ask for twin maintenance

WP3 develops learning components that (i) **update** uncertain product attributes from evidence and (ii) decide when to **ask** for additional information (inspection/sensing) versus proceeding conservatively or escalating. Learning-to-update uses calibrated models that can **abstain** when uncertain, producing conservative belief updates (e.g., probability-to-set conversion, conformal prediction sets, or guarded updates). Learning-to-ask uses **value-of-information** strategies constrained by admissible sensing actions, costs, and operational constraint envelopes.

Outputs. Conservative update models with justification metadata, information acquisition policies, and quantitative evaluation across uncertainty regimes against non-learning baselines.

WP4: Bounded semantic mediation: contract, gate, and formal assurance

WP4 designs the **constraint-preserving interface** between a semantic module (LLM) and holonic coordination. The semantic module is invoked only under defined exception triggers and returns recommendations in a **closed intent vocabulary** and schema. A **deterministic validation gate** enforces: (i) schema/type validity, (ii) grounding requirements against the twin trace, and critically (iii) **model-consistency checks** that compile intents into action candidates and intersect them with the **supervisor-enabled set** computed from the DES model. A structured **assurance case** links claims (invariant preservation) to evidence (proof sketches/containment arguments, gate checks, adversarial tests, runtime logs).

Outputs. Mediation contract and intent taxonomy, validation-gate implementation with adversarial test suite, and a formal containment argument with an invariant-preserving assurance case providing traceability to supervisory constraints.

WP5: Constraint-preservation and performance envelope, baselines, and transferable patterns

WP5 performs end-to-end characterization across uncertainty regimes and coordination design choices. We compare: (i) non-semantic holonic baselines, (ii) holonic + uncertainty-aware twin (WP2), (iii) holonic + learning-to-update/ask (WP3), and (iv) full BSM with gated semantic assistance (WP4). We quantify exception resolution, recovery dynamics, throughput, blocking/rework, inspection cost, escalation rate, and invariant adherence. We then conduct **sensitivity analysis** to map a constraint-preservation and performance envelope and distill **transferable integration patterns** and a portability checklist (what must be re-specified and re-verified when porting to a different cell or station configuration).

Outputs. Comparative evaluation results, envelope maps, reusable artifacts (schemas, validators, benchmark definitions), and deployment/porting guidance.

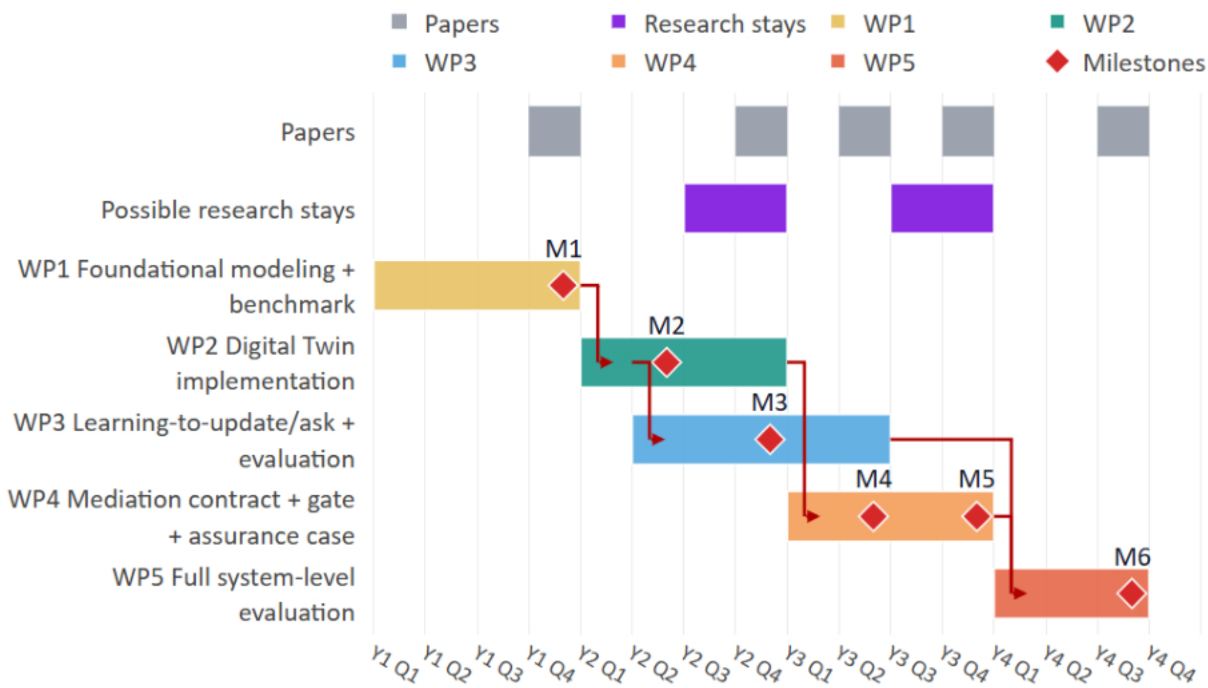

Figure 4. *Gantt chart of the four-year research plan, showing the temporal overlap and dependencies of WP1-WP5, associated milestone deliveries M1-M6, and planned paper outputs. Critical path in red. Any mobility period will remain within the limits set by FWO regulations.*

Progress monitoring, milestones, and critical path

Monitoring: Progress will be tracked through continuous technical integration, quarterly milestone reviews with explicit acceptance criteria, and reproducible experiment reports generated from the event-sourced logs.

Critical path: WP1 is foundational. WP2 depends on WP1. WP3 depends on WP2 and proceeds largely in parallel with WP4 once WP2 interfaces are stable. WP4 depends on WP1 and WP2. WP5 depends on the integrated stack (WP1-WP4, including WP3 learning components).

Milestones (with acceptance criteria):

M1 (end Y1): Reference cell + invariants + uncertainty regimes defined; evaluation platform reproduces baseline cell behavior under multiple regimes.

M2 (mid Y2): Event-sourced twin ingests multi-modal evidence; deterministic replay of full runs; UE5 twin can visualize traces and state.

M3 (end Y2): Conservative baseline belief updates validated; initial learning-to-update/ask prototypes run in simulation with documented calibration/abstention.

M4 (mid Y3): Mediation contract finalized; gate integrated; gate rejects invalid/adversarial outputs and only forwards supervisor-enabled decisions.

M5 (end Y3): Assurance case completed linking invariants to gate checks + tests + runtime logging; integrated system runs across core regimes.

M6 (end Y4): Envelope maps and patterns validated; public artifact package (as permitted) released; thesis-ready integrated narrative.

Scientific risks and contingency plans

Risk R1: Learning models are poorly calibrated or brittle under distribution shift.

Mitigation: prioritize conservative updates (abstention, conformal sets, guarded updates); retain rule-based baseline as fallback; use regime-based testing; enforce that learning can only shrink uncertainty when justified, otherwise defer to supervisor-enabled policies.

Risk R2: The validation gate becomes too restrictive, reducing semantic benefit.

Mitigation: iterate intent vocabulary and parameterization (still closed); add multi-step recommendations as *plans* that must be decomposed into supervisor-enabled single steps; evaluate benefit via controlled ablations; if needed, refocus semantic role on explanation/diagnosis while actions remain strictly gate-bounded.

Risk R3: Physical cell integration delays (hardware downtime, sensor instability).

Mitigation: simulation-first approach via WP1 evaluation environment and UE5 twin; keep experiments reproducible in simulation; validate on hardware opportunistically without blocking core contributions.

Risk R4: Formal verification complexity (state-space blow-up).

Mitigation: use compositional containment arguments and enabled-set intersection as the primary guarantee; restrict proof obligations to the mediation boundary; use testable contracts and runtime checks as additional evidence in the assurance case.

Inventiveness of the research methodology

The methodology is inventive in how it **integrates formal supervisory control, uncertainty-aware twins, conservative learning, and semantic assistance** into a single assurance story suitable for brownfield automation: (i) **Bounded semantic mediation by design:** semantic outputs are compiled and constrained to the **supervisor-enabled decision set**, enabling a strong “no new behavior” containment guarantee rather than heuristic filtering, (ii) **Event-sourced twin grounding for**

accountability: semantic and learning components are grounded in **replayable traces** with provenance and confidence metadata, supporting auditability and precise failure analysis, (iii) **Constraint-preserving learning under partial observability:** learning is used to reduce uncertainty and trigger information acquisition, but is structurally prevented from expanding admissible actions beyond formally enforced operational constraints.

Transparency, reproducibility, and research integrity measures

Version-controlled artifacts: models (DES/specs), schemas, coordination code, experiment scripts, and documentation maintained in structured repositories with tagged releases.

Reproducible experiments: all runs are driven by configuration files specifying regimes, seeds, policies, and model versions; results regenerated from the event store via replay.

Data and model lineage: event logs include evidence provenance, timestamps, sensor configuration, software versions, and model identifiers; trained models are stored with training configs and evaluation summaries.

Benchmark release strategy: release (i) regime definitions, (ii) simulator interfaces, (iii) contract/gate schemas and tests, and (iv) sanitized/synthetic traces where raw data cannot be shared; document any constraints clearly.

Predefined metrics and ablations: evaluation metrics and baseline comparisons are fixed early (WP1/WP5) to avoid retrospective metric selection.

Dissemination of research results (scientific/technical)

Dissemination will target peer-reviewed venues to share the project's reusable artifacts, distinct from public outreach, and will build on the group's track record in circular economy and demanufacturing. Publications will follow the work package logic: (i) benchmark and structural uncertainty modeling (WP1), (ii) event-sourced uncertainty-aware digital twin and trace layer (WP2), (iii) conservative learning-to-update/ask (WP3), (iv) bounded semantic mediation and assurance case (WP4), and (v) system-level envelope mapping and transferable patterns (WP5). Target journals include Journal of Cleaner Production, Resources, Conservation & Recycling, Waste Management, Journal of Industrial Ecology, and International Journal of Production Research, complemented where appropriate by manufacturing-automation and robotics journals. Target conferences and scientific communities include CIRP LCE (Life Cycle Engineering), Winter Simulation Conference (INFORMS), IEEE CASE, and IFAC MIM. Open-source artifacts will comprise benchmark regimes, schemas, validator/gate reference implementations, and containerized experiment pipelines to enable reuse. Community engagement will include workshops/special sessions on trustworthy AI for demanufacturing.

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PHD FELLOWSHIP STRATEGIC BASIC RESEARCH
APPLICATION POTENTIAL

Strategic dimension and application potential

Real-world challenge and relevance. Waste electrical and electronic equipment (WEEE) is increasing in scale and complexity, with significant losses of recoverable materials. Global e-waste reached 62 Mt in 2022 and is projected to 82 Mt by 2030, while only 22.3% is formally recycled under environmentally sound conditions [26]. In the EU, the WEEE collection rate was 37.5% in 2023 (in 2022: 40.6%), well below the 65% Directive target [27], indicating a persistent gap between generated and properly treated flows.
For Flanders and Belgium, this challenge is strategically relevant given high material import dependency and the ambition to reduce the material footprint of Flemish consumption by 30% by 2030 (Vision 2050) [29]. End-of-life electronics contain high-value and critical raw materials (rare earths as well), making improved reuse and recovery directly relevant for reducing external dependence. Belgium's established WEEE coordination structures (e.g., Recupel), combined with an active circular-economy policy framework and KU Leuven's industrially representative demanufacturing testbed at the "Re- and Demanufacturing Lab", provide a concrete basis to develop and validate coordination solutions within the Flemish circular manufacturing ecosystem.

How the research contributes and how insights translate. High-quality reuse and high-purity recovery require disassembly despite uncertain and variable product conditions, which limits automation in heterogeneous device streams. The project develops a decision-support architecture that improves auditable decision-making under uncertainty while preserving certified control constraints. It adds a controlled reasoning layer for exception handling within defined operational boundaries of existing systems. An evidence-linked digital twin maintains a time-stamped record of system state, uncertainty levels, and decision rationales. This supports consistent routing, transparent traceability, and more reliable reuse triage and selective disassembly, reducing avoidable misclassification and variability in recovery outcomes.
Beyond individual cells, the trace layer improves coordination with refurbishment and remanufacturing networks by making uncertainty explicit and downstream flows more predictable. While validated in demanufacturing, the underlying principle (traceable decision support under incomplete information without overriding certified control logic) is transferable to other industrial contexts such as field maintenance and hybrid human-robot repair operations.
The project will be strategically linked to the recently confirmed EU Horizon Europe project Euro-FMX: European TrUstworthy GeneRative AI & AutonOmisation Frontier Models for Manufacturing-X Competitiveness. Within this ecosystem, the proposed architecture contributes with a constraint-preserving, auditable decision-support layer for high-variability circular operations. The alignment ensures methodological exchange with frontier industrial AI developments while grounding generative and automation and autonomy concepts in certified, operationally constrained control environments.

Longer-term applications and concrete scale. In the longer term, the project delivers reusable architectural components for high-variability circular operations, including coordination models, trace schemas, and validated exception-handling patterns. Application domains include adaptive disassembly under uncertain product conditions (e.g., stripped screws, hidden fasteners), earlier identification of reuse-eligible units, constraint-compliant routing of battery-bearing devices, and audit-ready traceability across actors.

Sensitivity examples highlight leverage at system level. Using Recupel's 2024 baselines (134,446 t collected; about 6,350 t reused), a 5-10% relative increase in reuse capture corresponds to about 318-635 t/year, while a 1% relative improvement in recovery yield corresponds to about 1,344 t/year [28]. Even modest improvements in routing and triage thus translate into significant material gains, strengthening the case for deployment and scalability across facilities and streams.

Beyond demanufacturing, the same principle of traceable, bounded decision support under uncertainty is transferable to other high-variability industrial contexts, including advanced manufacturing sectors such as semiconductor production, where decision errors under incomplete information carry high cost and quality risks.

Sectors and actors who can use the results. The project is positioned as a cross-sector enabling research rather than a solution for a single stakeholder. Potential users include industrial actors first, such as battery and EPR organisations (e.g., BEBAT, with whom the co-promotor already collaborates), demanufacturing and refurbishment companies (e.g., TVH), electronics recyclers and pre-processors, and automation and machine-building companies integrating auditable decision support into hybrid human-robot cells, followed by public bodies monitoring circular-economy targets and compliance (e.g., OVAM, Vlaanderen Circulair) and broader collection-system and refurbishment/repair networks. The architectural components developed (bounded semantic mediation, validation gates, and trace-linked digital-twin schemas) constitute transferable software artifacts that can be integrated into automation and circular-technology platforms. In coordination with KU Leuven Research & Development, opportunities for intellectual property protection and post-project knowledge transfer will be explored where appropriate, including licensing, industrial research collaborations, or further maturation through KU Leuven Industrial Research Fund instruments.

Stakeholder engagement before, during, and after. Before project start, a structured needs-and-constraints mapping will be conducted with representatives from WEEE coordination bodies, recyclers, refurbishment actors, and automation providers to define representative exception scenarios and evaluation metrics, without reliance on a single partner.

During the project, engagement will be anchored by the PhD's industrial advisor and milestone-based consultations, including two validation workshops (mid-project and final year). The KU Leuven demonstrator will allow stakeholders to review logged runs and exception cases, and an anonymised exception library will ensure representativeness.

After the project, uptake will be supported through open technical artifacts (reference architectures, schemas, validation patterns, benchmark regimes), targeted workshops, and follow-up collaboration pathways via Flanders Make (in which both promotors are principal investigators) and relevant VLAIO instruments.

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